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Lab9

OOPS

Task1 :

```
Lab9.cpp
1  #include <iostream>
2  #include <string>
3  using namespace std;
4  class Robot
5  {
6  public:
7      string robotId;
8      string robotType;
9      int battery;
10     float speed;
11 };
12 int main()
13 {
14     Robot r1, r2, r3;
15     r1.robotId = "R-001";
16     r1.robotType = "Picker";
17     r1.battery = 90;
18     r1.speed = 1.5;
19     r2.robotId = "R-002";
20     r2.robotType = "Scout";
21     r2.battery = 55;
```

```
C:\Users\itx ninja\OneDrive\CD  + -
--- Robot Fleet Status ---
ID: R-001 | Type: Picker | Battery: 90% | Speed: 1.5 m/s
ID: R-002 | Type: Scout | Battery: 55% | Speed: 2.2 m/s
ID: R-003 | Type: Carrier | Battery: 20% | Speed: 0.8 m/s

-----
Process exited after 0.1953 seconds with return value 0
Press any key to continue . . . |
```

Task2:

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4  class Sensor
5  {
6  public:
7      string sensorId;
8      string sensorType;
9      float reading;
10     bool isActive;
11 };
12 int main()
13 {
14     Sensor s1, s2, s3;
15     s1.sensorId = "S-01";
16     s1.sensorType = "Ultrasonic";
17     s1.reading = 25.4;
18     s1.isActive = true;
19     s2.sensorId = "S-02";
20     s2.sensorType = "Infrared";
```

```
C:\Users\itx ninja\OneDrive\CD  + -
--- Sensor Status Report ---
ID: S-01 | Type: Ultrasonic | Reading: 25.4 | Active: true
ID: S-02 | Type: Infrared | Reading: 10.1 | Active: false
ID: S-03 | Type: Camera | Reading: 0 | Active: true

-----
Process exited after 0.1871 seconds with return value 0
Press any key to continue . . .
```

Hometask1:

```
#include <iostream>
#include <string>
using namespace std;
class AIModel
{
public:
    string modelName;
    string version;
    string algorithm;
    float accuracy;
    bool isDeployed;
};
int main()
{
    AIModel model1, model2, model3;
    model1.modelName = "ResNet-50";
    model1.version = "v2.1";
    model1.algorithm = "CNN";
    model1.accuracy = 0.91;
    model1.isDeployed = true;
    model2.modelName = "SVM-Classifer";
    model2.version = "v1.0";
    model2.algorithm = "SVM";
    model2.accuracy = 0.78;
    model2.isDeployed = false;
```

```
C:\Users\itx ninja\OneDrive\CD  + -
Name : SVM-Classifier
Version : v1.0
Algorithm : SVM
Accuracy : 0.78
Deployed : false
=== AI Model Card ===
Name : YOLOv5
Version : v3.0
Algorithm : CNN
Accuracy : 0.95
Deployed : true

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Process exited after 0.1912 seconds with return value 0
Press any key to continue . . . |
```